Retail Transaction Analysis Project Report

Abstract

This project analyzes online retail transactional data to identify profit-draining categories, optimize inventory turnover, and uncover seasonal product behavior. The dataset contains invoice-level product sales information, which is transformed and visualized using Python and Tableau to extract business insights. The analysis highlights top-performing products, slow movers, monthly revenue trends, and seasonal demand fluctuations, helping in data-driven decision-making for retail strategy and stock optimization.

Introduction

Retail businesses generate large volumes of transactional data every day. However, raw data alone does not provide insights until it is cleaned, processed, and analyzed. This project demonstrates how business analytics can convert transactional retail data into actionable insights. The objective is to understand profits and inventory performance while identifying hidden seasonal opportunities. The final outcome is presented as an interactive Tableau dashboard.

Tools Used

- **Python (Pandas, NumPy, Matplotlib, Seaborn)** – for data cleaning, processing, and feature engineering
- **SQL (via Python environment)** – for calculating profitability metrics
- **Tableau** – for building interactive dashboards and visualizations
- **CSV Dataset (Online Retail Kaggle Dataset)** – as the primary data source

Steps Involved

1. **Data Import & Cleaning**
2. Loaded CSV dataset into Python
3. Handled missing rows and invalid entries
4. Converted date fields into usable time dimensions such as Month, Year, and Season
5. **Feature Engineering**
6. Created Revenue column (Quantity × Price)
7. Added Month-Year and Seasonal classifications
8. Identified slow-moving and low-performing products
9. **Profitability & Inventory Analysis**
10. Used SQL-style grouping in Python to compute revenue per category/product
11. Calculated top-performing and low-performing products
12. **Seasonal Trend Analysis**
13. Extracted Month and Season fields

14. Created product demand patterns across seasons

15. **Visualization in Tableau**

16. Built multiple sheets such as monthly revenue trend, top products, seasonal product comparison, and inventory mapping

17. Added Season and Country filters and applied them across all sheets for interactivity

18. **Dashboard Creation**

19. Combined all visualizations into one dashboard
20. Linked filters globally
21. Enabled drill-down and comparative performance analysis

Dashboard Summary

1. **Monthly Revenue Trend (Sheet 1)** – Shows overall monthly performance and helps identify peak and low sales periods.
2. **Top Products by Revenue (Sheet 2)** – Highlights the highest revenue-generating products to focus on profitable stock.
3. **Profit-Draining Products (Sheet 3)** – Identifies items that reduce profitability due to low revenue but high stock movement.
4. **Slow-Moving Products (Sheet 4)** – Displays products with very low inventory turnover to help optimize stock planning.
5. **Hidden Seasonal Stars (Sheet 5)** – Reveals products with seasonal demand spikes that may not perform well annually but rise sharply during specific months.
6. **Inventory vs Revenue Heatmap (Sheet 6)** – Compares quantity sold with revenue to evaluate how efficiently inventory converts into revenue.

Conclusion

This project demonstrates the power of business analytics in retail. By transforming raw transactional data into insights, it identifies top contributors to revenue as well as products causing profit leakage. It also highlights seasonal demand opportunities that can support inventory planning and promotional strategies. The interactive Tableau dashboard makes it easy for business stakeholders to explore the data and identify trends, guiding better decision-making for profitability and stock optimization.